

(a) Shared prediction pipeline

Lifetime sample X
Sort and standardize

Three-output MLP
 $n \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 3$

Decode parameters
Shape, scale, location

Weibull formula
Reliability life points

Inference uses samples only. Truth supplies simulation labels and reference thresholds.

(b) Training and validation: separately trained procedures

P Parameter reference

Train: minimize parameter loss
Validate: select lowest parameter loss

Reference for parameter recovery

Q Target supervision ($R = 0.95$)

Train: minimize target life-point loss
Validate: select lowest target loss

Test target gains and cross-point costs

QCP Add parameter constraint

Train: target loss + constraint penalty
Validate: lowest target loss if feasible

Test retained gains and repair

QCP threshold: $1.5 \times$ earlier P reference validation loss; feasibility concerns the validation-set mean.

Controls | QP: fixed weighting · P_QSELECT: common validation · Q_FEAS: feasible selection · QMULTI: three-point loss